LESSON 5.4

Apply Area Concepts to Solve Problems

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a composite figure by decomposing it into basic shapes, including a regular polygon split into triangles, and adding or subtracting the areas.

WHY THIS MATTERS Your architecture firm is calculating the floor area of an L-shaped room for a client's new home. The contractor needs the total area to order the correct amount of flooring. The room can be split into two rectangles: one is 12 feet by 8 feet and the other is 6 feet by 5 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Composite Figure *Spanish: Figura compuesta*	A shape made by putting two or more simple shapes together.	triangles joined together
Decompose *Spanish: Descomponer*	To break a shape into smaller, simpler shapes.	hexagon → 6 triangles
Add *Spanish: Sumar*	Add up the areas of the smaller shapes to get the total.	T-shaped hallway: top rectangle = 30 sq ft, bottom rectangle = 28 sq ft → total = 30 + 28 = 58 sq ft
Subtract *Spanish: Restar*	Take away the area of a missing piece from a bigger shape.	A 14×10 pool with a 6×4 cutout: 140 − 24 = 116 sq ft of water surface
Formula *Spanish: Fórmula*	A math rule written with symbols.	Rectangle: A = l × w; Triangle: A = ½ × b × h; use the right formula for each piece of a composite figure
Regular polygon *Spanish: Polígono regular*	A shape whose sides are all the same length and whose angles are all the same size.	stop sign, hexagon tile

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Regular pentagon: sides 10 cm, apothem about 7 cm.

1. 5 sides make 5 triangles.
2. One triangle: ½ × 10 × 7 = 35 cm².
3. Five triangles: 5 × 35 = 175.

Answer: Area ≈ 175 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now an L-shape made of a 5 ft by 4 ft rectangle and a 3 ft by 2 ft rectangle.

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I decomposed the figure into _____ congruent _____ .
- The area of one triangle is _____ square units.
- There are _____ triangles, so the total area is _____ .
- I know the triangles are congruent because _____ .

Word bank regular polygon | decompose | congruent | triangle | center | base | height | multiply | total | area | square units | composite figure

WATCH OUT!

A common mistake

A common mistake in Area of Composite Figures is adding every piece even when a piece has been cut out of the figure. For a 10 ft by 8 ft rectangle with a 3 ft by 2 ft rectangular notch cut from one corner, students compute the full rectangle 10 × 8 = 80 sq ft and stop, or they add the notch instead of removing it (80 + 6 = 86 sq ft) instead of subtracting it: 80 − (3 × 2) = 80 − 6 = 74 sq ft. Before submitting, check whether each piece is part of the figure (add it) or missing from the figure (subtract it).

LESSON 5.4

Apply Area Concepts to Solve Problems

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each composite figure by whether you ADD or SUBTRACT the areas to find the total.

Groups: Add the areas | Subtract the areas

_____ L-shaped hallway (two rectangles joined)

_____ Wall with a window cut out

_____ T-shaped stage (two rectangles joined)

_____ Floor with a trapdoor removed

_____ U-shaped pool (rectangle with center removed)

_____ Plus-sign shape (rectangles joined)

HINT Look for action words on each card: 'joined' versus 'cut out' or 'removed'.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

An L-shaped room is made of two rectangles: 10 ft × 6 ft and 4 ft × 3 ft. What is the total area?

(A) 12 sq ft
(B) 60 sq ft
(C) 72 sq ft
(D) 72 ft

HINT The L-shape is already split into two rectangles for you: 10 × 6 and 4 × 3.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A rectangular patio is 15 ft × 10 ft with a 5 ft × 4 ft rectangular flower bed cut out. What is the remaining area?

(A) 20 sq ft
(B) 130 sq ft
(C) 150 sq ft
(D) 170 sq ft

HINT The flower bed is CUT OUT of the patio — that phrase tells you which operation to use.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

(A) 8 sq units
(B) 15 sq units
(C) 20 sq units
(D) 23 sq units

HINT Two rectangles make this L: one is 5 × 3 and the other is 2 × 4.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

When should you subtract areas instead of adding them to find the area of a composite figure?

(A) When the shapes overlap
(B) When one shape is larger than the other
(C) Always subtract — never add
(D) When a piece is cut out or removed from a larger shape

HINT Think of real examples: a window in a wall versus two rooms joined into an L-shape.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Identify the shapes** An L-shaped room is made of two joined rectangles: 8 ft × 5 ft and 4 ft × 3 ft.
2. **Find each area** First rectangle = 8 × 5 = 40 sq ft. Second rectangle = 4 × 3 = 12 sq ft.
3. **Combine the areas** Total = 40 × 12 = 480 sq ft.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Look at the last step. When two pieces are joined, do you add or multiply their areas?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

For the U-shaped pool, why did we subtract instead of add? When do you add areas and when do you subtract?

SENTENCE STARTER I ___ the areas because the composite figure is made by ___ a piece from a larger shape, so the total area is ___ - ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.4

Apply Area Concepts to Solve Problems

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Figure	Shape 1 Area	Shape 2 Area	Operation	Total Area
L-shaped deck	48 sq ft	20 sq ft	Add	
Floor with alcove	90 sq ft	15 sq ft	Add	
Wall with window cutout	120 sq ft	18 sq ft	Subtract	
Courtyard with fountain cutout	200 sq ft	36 sq ft	Subtract	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each composite figure to its total area.

_____ L-shape: 10×6 + 4×3

_____ T-shape: 14×4 + 6×10

_____ Wall 15×10 minus window 5×4

_____ Pool 14×10 minus cutout 6×5

_____ L-shape: 8×5 + 3×6

_____ Floor 12×8 minus vent 2×3

(A) 72 sq ft

(B) 116 sq ft

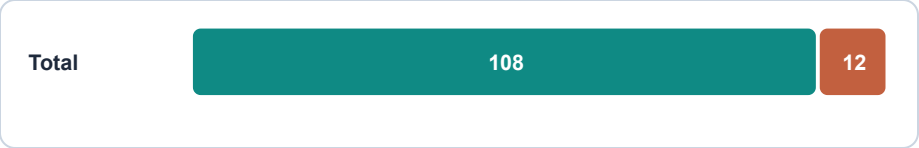
(C) 130 sq ft

(D) 110 sq ft

(E) 58 sq ft

(F) 90 sq ft

3 **Use the model to solve. Show your work.**



Wall area = 120 sq ft → painted area (108) + window cutout (12)

SHOW YOUR WORK

Answer: _____

4 **Write the name of the correct group on each line.**

Calculate each composite figure's area. Sort by whether the area is more than 100 sq ft or not.

Groups: **Area > 100 sq ft** **Area ≤ 100 sq ft**

_____ L-shape: 12×8 + 6×5

_____ T-shape: 10×3 + 4×7

_____ Wall 15×10 − window 5×4

_____ L-shape: 9×7 + 3×4

_____ Floor 20×8 − vent 6×3

_____ U-shape: 10×8 − 4×3

PART 2 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Identify the shapes** A 16 ft × 9 ft floor has a 4 ft × 5 ft supply closet cut out of one corner. Find the usable floor area.
2. **Find the large rectangle area** A = 16 × 9 = 144 sq ft.
3. **Find the cutout area** A = 4 × 5 = 20 sq ft.
4. **Find the usable area** Usable area = 144 + 20 = 164 sq ft.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

For the U-shaped pool, why did we subtract instead of add? When do you add areas and when do you subtract?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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Challenge · Extension

MASTERY CHECK
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

This composite figure is a house shape: a rectangle topped with a triangle. Find each area and the total.

Part	Shape	Dimensions	Area
Walls	Rectangle	10 ft × 8 ft	
Roof	Triangle	b = 10 ft, h = 4 ft	
Total	—	—	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Area Error

1. **Identify the shapes** Rectangle: 16 ft × 10 ft, with a 4 ft × 6 ft piece cut from the corner
2. **Find the large rectangle area** A = 16 × 10 = 160 sq ft
3. **Find the cutout area** A = 4 × 6 = 24 sq ft
4. **Find total area** Total = 160 + 24 = 184 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A T-shaped room can be decomposed in two different ways. Show both ways and prove they give the same total area. Use a T-shape where the top is 14 ft × 4 ft and the stem is 6 ft × 10 ft.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

For the U-shaped pool, why did we subtract instead of add? When do you add areas and when do you subtract?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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